Determination of the gravity vector

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I. Introduction

Initially, I discovered that the light on a traffic baton often causes misunderstanding and miscommunication since it is hard to understand the direction the policeman is pointing at, which potentially leads to accidents or fatalities. **Fig.1** illustrates a man waving a baton. Traditionally, when a conventional baton is used, the light is always on. However, it is hard to distinguish the waving direction when the light is always on. By observing these phenomena, I came up with an idea to improve the design of the traffic baton. Therefore, my project is to develop a baton that can automatically adjust its light based on the waving direction. Referring to **Fig. 1**, suppose the waving direction is from point A to point B. The improved baton will have its lights on in the waving trajectory, S1. In contrast, the improved baton will have its lights off in the waving trajectory, S2. In this way, the waving direction can be clearly indicated for pedestrians or drivers. The motion of the improved baton can be seen from the video links in the **Bibliography**. To accomplish this task, it is crucial to determine the state of the baton in order to control its light on or off. The resulting light-on and -off can clearly indicate the desired direction for the pedestrians, which avoids unnecessary accidents. For determining the state of the baton, it is straightforward to equip the baton with an accelerometer. However, the accelerometer senses not only the baton's acceleration but also the gravity vector. To obtain the acceleration of the baton, the accelerometer's output needs to be subtracted by the gravity vector. The gravity vector is necessary to obtain the baton's acceleration from the accelerometer's measurement. It is crucial to determine the state of the baton, so to implement the improved baton. Therefore, I came up with a research question of "*How to identify the gravity vector using the measured three-dimensional acceleration of a moving object in real-time?*" for the fundamental research that contributes to the improvement of the design of the baton.

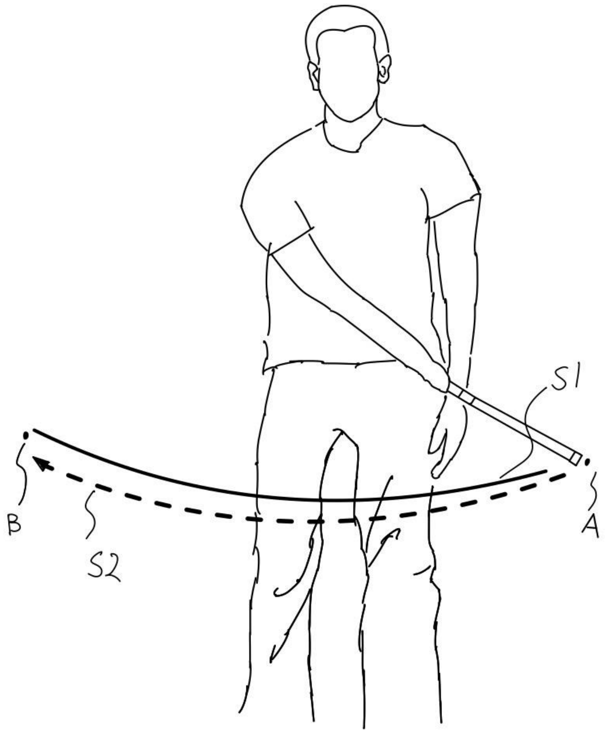


Fig. 1 Demonstration of baton motion

Sometimes, an accelerometer is combined with a gyroscope in an integrated circuit. In this study, only an accelerometer is required to estimate the gravity vector. Specifically, a MATLAB simulation of a tri-axial accelerometer's output is used to identify the gravity vector in a body-fixed frame, enabling the estimation of an object's tilt and acceleration. This study will focus on the comparison of the gravitational acceleration and the object's linear accelerations between the proposed method and a conventional method, the sensor reading method, to illustrate the feasibility of the methodology and to demonstrate the advantages of the proposed scheme over the conventional method.

This study is further explained as follows. Section II provides background materials, including necessary mathematical concepts. Section III describes the methodology that outlines the variables and the approach of this study. Subsequently, Section IV is the data collection that shows the results simulated in MATLAB. Section V is the data analysis that presents the analyzed results referring to the research question. Following this, Section VI is the evaluation that discusses the limitation and weakness of this study. Finally, conclusions are given in Section VII.

II. Background

Understanding the motion and orientation of a body in three-dimensional space is vital. The Background section provides context for the study by outlining the research problem.

A. Definition of Coordinate Systems

The coordinate system of interest is tri-axial, and three axes, x-, y-, and z-axes form a right-handed coordinate system. For orientation description, three rotation angles, referred to as Euler angles, are denoted as roll ϕ, pitch θ, and yaw ψ around the x-, y-, and z-axes, respectively. These angles are illustrated in **Fig. 2**.

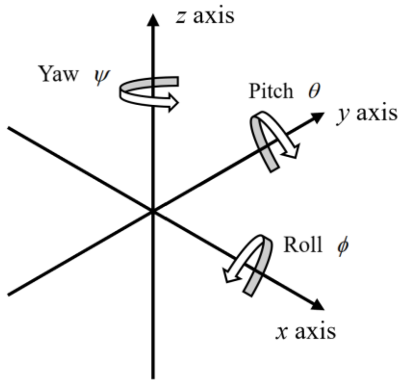


Fig. 2 Definition of a Coordinate System

An inertial frame is a coordinate system, in which Newton's law of motion is valid (DiSalle, R., 2002). The inertial frame is not accelerating or rotating. In order to describe the orientation of a body, another coordinate system is attached to the body (Craig, J. J., 2005). The x-axis is aligned along the body axis. The z-axis points up. The y-axis is aligned perpendicular to the x- and z-axes, giving a right-handed coordinate system. This coordinate system is referred to as a body-fixed frame.

B. Rotation Matrices from Inertial Frame to Body-Fixed Frame

In the inertial frame, the gravity vector is a constant vector pointing to the negative z-axis. That is, the z-axis is parallel to the gravity vector. In contrast, the body-fixed frame is a coordinate system that is attached to a specific movable object. When the object is in motion, the gravity vector can be time-varying in the body-fixed frame; that is, the gravity vector is a function of time.

A sequence of 3 rotations can correlate any two coordinate systems. The roll ϕ, pitch θ, and yaw ψ rotation matrices of the inertial frame can be transformed into the body-fixed frame. There are six possible sequence combinations in the order of the three-rotation axes. Different combinations of sequences in the order of rotating axes lead to a different matrix expression. That is, rotations do not commute. In principle, all are equally valid. However, this study only considers one specific sequence of rotations, which is from the rotation of the *z*-axis to the *y*-axis, and then to the *x*-axis. It can be denoted as $\mathbf{R}_{xyz}$: (Pedley, M., 2013)

$$\mathbf{R}_{xyz} = \mathbf{R}_x(\phi)\mathbf{R}_y(\theta)\mathbf{R}_z(\psi)$$
$$= \begin{bmatrix} \cos\theta\cos\psi & \cos\theta\sin\psi & -\sin\theta \\ \cos\psi\sin\theta\sin\phi - \cos\phi\sin\psi & \cos\phi\cos\psi + \sin\theta\sin\phi\sin\psi & \cos\theta\sin\phi \\ \cos\phi\cos\psi\sin\theta + \sin\phi\sin\psi & \cos\phi\sin\theta\sin\psi - \cos\psi\sin\phi & \cos\theta\cos\phi \end{bmatrix}, \tag{1}$$

in which

$$\mathbf{R}_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix}, \tag{2}$$

$$\mathbf{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}, \tag{3}$$

$$\mathbf{R}_z(\psi) = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{4}$$

Let the gravity vector in the body-fixed frame be denoted by $\mathbf{g} = \begin{bmatrix} g_x & g_y & g_z \end{bmatrix}^{\mathrm{T}}$, in which g_x, g_y, and g_z are the *x*-, *y*-, and *z*-axis components of the gravity vector in the body-fixed frame. By only considering the rotation sequence $\mathbf{R}_{xyz}$, the transformation of the gravity vector in the inertial frame to the body-fixed frame can be realized using equation (1), giving:

$$\mathbf{g} = \begin{bmatrix} g_x \\ g_y \\ g_z \end{bmatrix} = \mathbf{R}_{xyz} \begin{bmatrix} 0 \\ 0 \\ -\mathrm{g}_0 \end{bmatrix} = -\mathrm{g}_0 \begin{bmatrix} -\sin\theta \\ \cos\theta\sin\phi \\ \cos\theta\cos\phi \end{bmatrix}, \tag{5}$$

in which g_0 denotes the standard acceleration of gravity and equals 9.80665 m/s^2 by standard. From equation (5), the roll and pitch angles can be solved. Note that only the rotation sequences of $\mathbf{R}_{xyz}$ and $\mathbf{R}_{yxz}$ can be applied as being suitable for determining the orientation. The first rotation is in yaw ψ around the z-axis that is initially aligned with the gravitational field pointing downwards. The rotation sequences of $\mathbf{R}_{xyz}$ and $\mathbf{R}_{yxz}$ can then eliminate the yaw rotation ψ in the matrices and solve for the unique solution for the roll ϕ and pitch θ.

C. Accelerometer Readings

An accelerometer's output contains the gravity vector and the translational acceleration of a moving object, on which the accelerometer is mounted. Denote the accelerometer's output as $\mathbf{a} = \begin{bmatrix} a_x & a_y & a_z \end{bmatrix}^{\mathrm{T}}$, in which a_x, a_y, and a_z are respectively the accelerometer outputs in the x-, y-, and z-axes. The accelerometer's output is a combination of three terms: (Luinge, H.J. & Veltink, P.H., 2004)

$$\mathbf{a} = \mathbf{g} + \mathbf{l} + \mathbf{n}, \tag{6}$$

in which $\mathbf{l}$ denotes the linear acceleration of the object, and $\mathbf{n}$ denotes measurement noise. It should be remarked that, if the linear acceleration of the object is small, then the accelerometer's output directly reflects the gravity vector. Unfortunately, this is usually not the case as the accelerometer is attached to a moving object.

D. Applications of an Estimated Gravity Vector

One application of an estimated gravity vector is to determine the inclination of an object (Hemminki, S., Nurmi, P., & Tarkoma, S., 2014). Denote an estimate of $\mathbf{g}$ as $\hat{\mathbf{g}} = \begin{bmatrix} \hat{g}_x & \hat{g}_y & \hat{g}_z \end{bmatrix}^{\mathrm{T}}$, in which $\hat{g}_x$, $\hat{g}_y$, and $\hat{g}_z$ are respectively the estimates of g_x, g_y, and g_z. From equation (5), the roll and pitch angles can be estimated by:

$$\hat{\phi} = \tan^{-1}\left(\frac{\hat{g}_y}{\hat{g}_z}\right), \qquad \hat{\theta} = -\tan^{-1}\left(\frac{\hat{g}_x}{\sqrt{\hat{g}_y^2 + \hat{g}_z^2}}\right). \tag{7}$$

The other application of the estimated gravity vector is to determine the linear acceleration of an object. The estimate of the object's linear acceleration in the body-fixed frame can be given by $\hat{\mathbf{l}} = \mathbf{a} - \hat{\mathbf{g}} = \begin{bmatrix} a_x & a_y & a_z \end{bmatrix}^{\mathrm{T}} - \begin{bmatrix} \hat{g}_x & \hat{g}_y & \hat{g}_z \end{bmatrix}^{\mathrm{T}}$.

E. Conventional Method of Estimating the Gravity Vector

In the conventional method of estimating the tilt of an object, a 3D accelerometer is used (Pedley, M., 2013; Manjiyani et al., 2018; Luczak et al., 2006). In this method, the output of the 3D accelerometer is considered the gravity measurements. Therefore, the roll and pitch angles can be estimated by:

$$\hat{\phi} = \tan^{-1}\left(\frac{a_y}{a_z}\right), \qquad \hat{\theta} = -\tan^{-1}\left(\frac{a_x}{\sqrt{a_y^2 + a_z^2}}\right), \tag{8}$$

where a_x, a_y, and a_z are the accelerometer's outputs in *x*-, *y*-, and *z*-directions respectively. However, when the object has significant linear acceleration, the accelerometer's outputs do not represent the gravity components. This conventional method is referred to as the sensor reading method in subsequent content. This study also uses only a 3D accelerometer, which only provides accelerations in three orthogonal directions, is used to identify the gravity vector in the body-fixed frame. However, this study proposes an algorithm for estimating the gravity vector from the 3D accelerometer's outputs while minimizing the influence of the object's linear acceleration.

III. Methodology

The following section provides a detailed explanation of the methodology in this study, including the research design, data collection processes, and analytical techniques utilized to address the research question.

Variables:

Independent Variable: frequency ω_l, and amplitude A_r.

Control Variable: k_e, k_g.

Dependent Variable: Tilt estimate; Estimate of the gravity vector; Estimate of the linear acceleration.

This work aims to estimate $\mathbf{g}$ based on the information on $\mathbf{a}$. The following assumption is made.

Assumption: The derivative of $\mathbf{g}$ with respect to time is zero, that is, $\dot{\mathbf{g}} = \mathbf{0}$.
This assumption implies that the time variation of $\mathbf{g}$ is small. Moreover, it also means that the linear acceleration of the object, $\mathbf{l}$, and the measurement noise, $\mathbf{n}$, should not contain static/quasi-static components. The gravity vector, $\mathbf{g}$, can also be obtained by transforming the constant vector, $\begin{bmatrix} 0 & 0 & \mathrm{g}_0 \end{bmatrix}^{\mathrm{T}}$, by a transformation matrix. This transformation matrix involves the sin/cos functions of pitch, roll, and yaw angles of the moving object in the inertial frame. Besides complex calculations, it is difficult to obtain these angles through the accelerometer's output. Sensor fusion with other sensors is usually used to estimate these angles well. In this work, the estimation of the gravity vector is done in the body-fixed frame, avoiding the complex transformation and the requirements of pitch and roll angles in the inertial frame.

Let g_x, g_y, and g_z denote the *x*-, *y*-, and *z*-axis components of the gravity vector in the body-fixed frame. That is, the gravity vector in the body-fixed frame is $\mathbf{g} = \begin{bmatrix} g_x & g_y & g_z \end{bmatrix}^{\mathrm{T}}$, which is time-varying. On the other hand, the gravity vector in the inertial frame is a constant vector, $\begin{bmatrix} 0 & 0 & \mathrm{g}_0 \end{bmatrix}^{\mathrm{T}}$. Both vectors have the same magnitude. Because the transformation (5) only rotates the gravity vector and does not alter the vector's magnitude. Based on this, the constraint on $\mathbf{g}$ is given by:

$$g_x^2 + g_y^2 + g_z^2 = \mathrm{g}_0^2, \text{ or } \left\|\mathbf{g}\right\|^2 = \mathrm{g}_0^2. \tag{9}$$

Based on the constraint on the magnitude of $\mathbf{g}$, define an error variable as:

$$e = \sqrt{\hat{g}_x^2 + \hat{g}_y^2 + \hat{g}_z^2} - \mathrm{g}_0 = \left\|\hat{\mathbf{g}}\right\| - \mathrm{g}_0. \tag{10}$$

To derive an estimation algorithm, let's define a quadratic function, *V*, by:

$$2V = k_e e^2 + k_g \left[(\hat{g}_x - g_x)^2 + (\hat{g}_y - g_y)^2 + (\hat{g}_z - g_z)^2 \right], \tag{11}$$

in which k_e and k_g are positive constants. The constants, k_e and k_g, are chosen to be positive so that the quadratic form (11) is positive definite. Taking the derivative of V with respect to time gives:

$$
\begin{aligned}
\dot{V} &= k_e \|\hat{\mathbf{g}}\|^{-1} e\left(\hat{g}_x\dot{\hat{g}}_x + \hat{g}_y\dot{\hat{g}}_y + \hat{g}_z\dot{\hat{g}}_z\right) + k_g\left[(\hat{g}_x - g_x)\dot{\hat{g}}_x + (\hat{g}_y - g_y)\dot{\hat{g}}_y + (\hat{g}_z - g_z)\dot{\hat{g}}_z\right] \\
&= \left[k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_x + k_g(\hat{g}_x - g_x)\right]\dot{\hat{g}}_x + \left[k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_y + k_g(\hat{g}_y - g_y)\right]\dot{\hat{g}}_y \\
&\quad + \left[k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_z + k_g(\hat{g}_z - g_z)\right]\dot{\hat{g}}_z,
\end{aligned} \tag{12}
$$

in which the assumption that the time derivatives of g_x, g_y, and g_z are zero is employed. To let $\dot{V}$ negative, the following estimation law is used:

$$
\begin{aligned}
\dot{\hat{g}}_x &= -k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_x - k_g(\hat{g}_x - g_x), \\
\dot{\hat{g}}_y &= -k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_y - k_g(\hat{g}_y - g_y), \\
\dot{\hat{g}}_z &= -k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_z - k_g(\hat{g}_z - g_z).
\end{aligned} \tag{13}
$$

Substituting (13) into (12) gives

$$
\begin{aligned}
\dot{V} = &-\left[k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_x + k_g(\hat{g}_x - g_x)\right]^2 - \left[k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_y + k_g(\hat{g}_y - g_y)\right]^2 \\
&-\left[k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_z + k_g(\hat{g}_z - g_z)\right]^2.
\end{aligned} \tag{14}
$$

The quadratic form (14) is negative semi-definite. Since $\dot{V}$ is negative semi-definite, it is concluded that $\hat{g}_x$, $\hat{g}_y$, and $\hat{g}_z$ are bounded. But the negative semi-definiteness of $\dot{V}$ cannot guarantee that $\hat{\mathbf{g}}$ approaches $\mathbf{g}$. To investigate the existence of convergence to zero as time approaches to infinity, note that $\dot{V} = 0$ implies that:

$$
\begin{aligned}
&k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_x + k_g(\hat{g}_x - g_x) = 0, \\
&k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_y + k_g(\hat{g}_y - g_y) = 0, \\
&k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_z + k_g(\hat{g}_z - g_z) = 0.
\end{aligned} \tag{15}
$$

Define the set, S:

$$
\begin{aligned}
S = \Big\{&(\hat{g}_x, \hat{g}_y, \hat{g}_z):\ k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_x + k_g(\hat{g}_x - g_x) = 0, \\
&k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_y + k_g(\hat{g}_y - g_y) = 0,\ k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_z + k_g(\hat{g}_z - g_z) = 0\Big\}.
\end{aligned} \tag{16}
$$

For this set, one has:

$$
\hat{g}_x = \frac{g_x}{1 + k_g^{-1}k_e \|\hat{\mathbf{g}}\|^{-1} e},\ \hat{g}_y = \frac{g_y}{1 + k_g^{-1}k_e \|\hat{\mathbf{g}}\|^{-1} e},\ \text{and}\ \hat{g}_z = \frac{g_z}{1 + k_g^{-1}k_e \|\hat{\mathbf{g}}\|^{-1} e}, \tag{17}
$$

giving:

$$\|\hat{\mathbf{g}}\|^2 = \hat{g}_x^2 + \hat{g}_y^2 + \hat{g}_z^2 = \frac{g_x^2 + g_y^2 + g_z^2}{\left(1 + k_g^{-1} k_e \|\hat{\mathbf{g}}\|^{-1} e\right)^2} = \frac{\mathrm{g}_0^2}{\left[1 + k_g^{-1} k_e \|\hat{\mathbf{g}}\|^{-1} \left(\|\hat{\mathbf{g}}\| - \mathrm{g}_0\right)\right]^2}. \quad (18)$$

In **Appendix A**, equation (18) is solved, which gives a unique solution of $\|\hat{\mathbf{g}}\| = \mathrm{g}_0$, that is, $e = 0$. Moreover, it is concluded that $\hat{\mathbf{g}}$ approaches $\mathbf{g}$ asymptotically, which means that the estimate of the gravity vector converges to the actual gravity vector as time approaches to infinity.

Since g_x, g_y, and g_z are usually unknown, the estimation law (13) cannot be implemented. Despite this, the estimation law (13) can act as a nonlinear filter when $\mathbf{g}$ is available. This nonlinear filter can be used to attenuate noises of the measured $\mathbf{g}$ while considering the magnitude constraint on $\mathbf{g}$. To be realizable in general situations, the estimation law is modified to:

$$\begin{aligned} \dot{\hat{g}}_x &= -k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_x - k_g(\hat{g}_x - a_x), \\ \dot{\hat{g}}_y &= -k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_y - k_g(\hat{g}_y - a_y), \\ \dot{\hat{g}}_z &= -k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_z - k_g(\hat{g}_z - a_z). \end{aligned} \quad (19)$$

in which $k_g > k_e > 0$ that is derived in **Appendix A**. There are two kinds of correction terms in equation (19). One is $-k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_\mathrm{i}$ for $\mathrm{i} = x, y, z,$ which are to satisfy the constraint on the sum of squares of $\hat{g}_\mathrm{i}$ for $\mathrm{i} = x, y, z.$ The other is $-k_g(\hat{g}_\mathrm{i} - a_\mathrm{i})$ for $\mathrm{i} = x, y, z.$ These correction terms are to lowpass-filter the accelerometer's outputs, a_i for $\mathrm{i} = x, y, z.$ Therefore, the estimate of the gravity vector is basically the static/quasi-static components of the accelerometer's outputs. This assumes that the gravity vector is static/quasi-static whereas the object's linear acceleration is dynamic and more time-varying.

Remark: If a tri-axial gyroscope is available, its output can be integrated into the proposed estimation law (19), which is described in **Appendix B**.

IV. Data Collection

In this study, the data collection is performed through the simulation in MATLAB. Let the orientation of an object be described by:

$$\begin{aligned}
&\phi = A_r \sin(2\pi \times 0.1 \times t), \\
&\theta = A_r \cos(2\pi \times 0.1 \times t), \qquad (20) \\
&\psi = A_r \sin(2\pi \times 0.1 \times t),
\end{aligned}$$

in which the amplitude, A_r, is an independent variable that determines the rotation amplitude. Moreover, let the object's linear acceleration, $\mathbf{l} = \begin{bmatrix} l_x & l_y & l_z \end{bmatrix}^{\mathrm{T}}$, be given by:

$$\begin{aligned}
&l_x = 5\sin(2\pi \times \omega_l \times t), \\
&l_y = 5\sin(2\pi \times \omega_l \times t + 2\pi/3), \qquad (21) \\
&l_z = 5\sin(2\pi \times \omega_l \times t + 4\pi/3),
\end{aligned}$$

in which the frequency, ω_l, is the other independent variable. **Table I** shows the classification of simulation cases. In this table, two rotation amplitudes and three frequencies of linear acceleration are chosen to test if the rotation amplitude and the frequency of linear acceleration will influence the effectiveness of the estimation law.

Table I. Classification of simulation cases

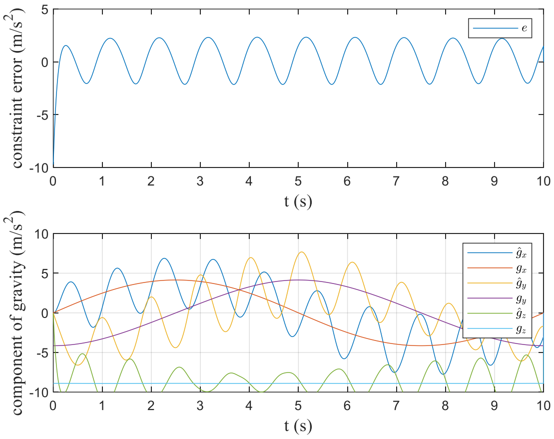
	1: $\omega_l = 1$ (rad/s)	2: $\omega_l = 5$ (rad/s)	3: $\omega_l = 10$ (rad/s)
A: $A_r = 25^\circ$	Case A1	Case A2	Case A3
B: $A_r = 50^\circ$	Case B1	Case B2	Case B3

The sensor reading method directly uses the accelerometer's outputs to be the $\mathbf{g}$, in which $\hat{g}_x = a_x$, $\hat{g}_y = a_y$, $\hat{g}_z = a_z$. In the sensor reading method, the roll and pitch angles are directly determined by the accelerometer's outputs, that is,

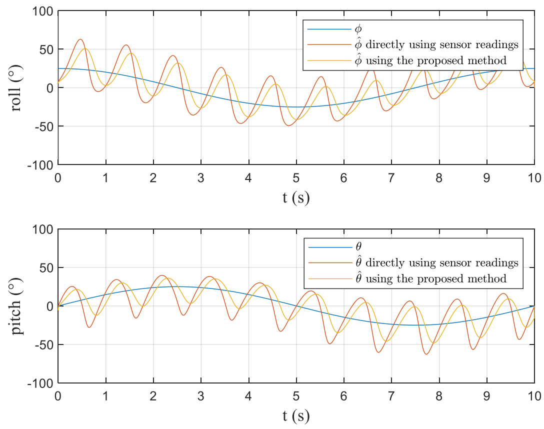
$$\hat{\phi} = \tan^{-1}\left(\frac{a_y}{a_z}\right), \qquad (22)$$

$$\hat{\theta} = -\tan^{-1}\left(\frac{\hat{g}_x}{\sqrt{\hat{g}_y^2 + \hat{g}_z^2}}\right). \tag{23}$$

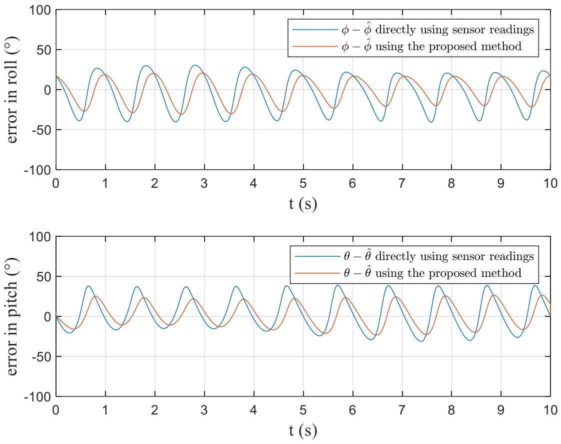
The proposed method uses (19) to estimate the $\mathbf{g}$ when k_e =5.9 and k_g =6, which satisfies the constraint, $k_g > k_e > 0$, but is determined by trial and error. According to (7) and (8), the $\hat{\theta}$ and $\hat{\phi}$ can be determined. **Fig. 3** illustrates the result for **Case A1**. In **Fig. 3** (c), the errors in angle are defined as the original angle subtracted by the estimated angle, which is $\theta - \hat{\theta}$ and $\phi - \hat{\phi}$. In this data collection section, only **Case A1** is illustrated in **Fig. 3** due to the assessment's page limit. The upper subplot of **Fig. 3** (a) shows the time evolution of the error variable defined in equation (10). The ideal value of the variables shown in **Fig. 3** (c) and the lower subplot of **Fig. 3** (d) is zero.



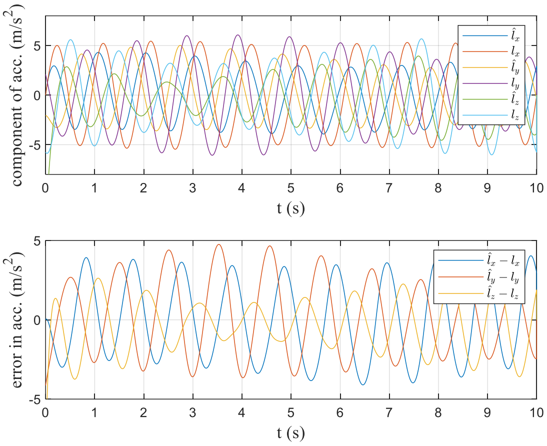
(a)



(b)



(c)



(d)

Fig. 3 Estimation with $A_r = 25^\circ$ and $\omega_l = 5$ (rad/s) in **Case A1**. (a) Components of the gravity vector; (b) Tilt estimation; (c) Error in tilt estimation; (d) Linear acceleration.

V. Data Analysis

To evaluate the performance of estimation, define the root mean square (*rms*) as the square root of the mean square value, which can be written as: (Hodson, T. O., 2022)

$$rms = \sqrt{\frac{1}{T}\int_0^T \left(f\left(t\right)\right)^2 dt}, \tag{24}$$

in which $f(t)$ denotes a function of time, T is defined as the definite time, T is 10 s in the simulation, and 0 and T define the boundary of the definite integral. **Table II** presents the comparison of the angle errors between the sensor reading method and the proposed method, in which the angle error is the difference between the estimated angle and the actual angle.

Table II. The *rms* value of ϕ error and θ error

	rms of ϕ error (°)		*rms* of θ error (°)	
	Sensor reading method	Proposed method	Sensor reading method	Proposed method
Case A1	22.9053	15.8363	21.4321	14.6363
Case A2	22.9064	4.5177	21.4215	4.2688
Case A3	22.9065	2.8265	21.4203	2.6323
Case B1	27.6861	18.2398	21.8269	15.2086
Case B2	27.7071	6.1455	21.7991	5.2346
Case B3	27.7089	4.5725	21.7965	4.0173

Comparing the results of the two methods from **Table II**, it is clear that both the *rms* of ϕ error and the *rms* of θ error of the sensor reading method are larger than those of the proposed method in every case. Therefore, the results show that it is beneficial to use the proposed method to estimate $\mathbf{g}$. In addition, the increase in frequency, ω_l, leads to a decrease in angle errors in the proposed method. However, the larger the value of the amplitude, A_r, the larger the angle errors. Therefore, to minimize the angle errors, the condition should be in high frequency, ω_l, and low amplitude, A_r.

In the proposed method, the constraint on the magnitude of the gravity vector is considered in the estimation law (19). To investigate the effect of incorporating this gravity constraint into the estimation law, the method without considering the gravity constraint is implemented in the simulation study by directly letting $k_e = 0,$ giving

$$\dot{\hat{g}}_x = -k_g(\hat{g}_x - a_x),\ \dot{\hat{g}}_y = -k_g(\hat{g}_y - a_y),\ \dot{\hat{g}}_z = -k_g(\hat{g}_z - a_z). \tag{25}$$

In the method without considering the gravity constraint, the gravity estimation law produces a gravity estimate that is simply a lowpass-filtered version of the accelerometer's output. Table III presents the *rms* values of gravity error and linear acceleration error for all the cases, in which the method without considering the gravity constraint is described by equation (25). The proposed method is simulated in the same conditions mentioned above.

Table III. The *rms* value of gravity error and linear acceleration error

	rms of gravity error (m/s^2)		*rms* of linear acceleration error (m/s^2)	
	Method w/o considering the gravity constraint	Proposed method	Method w/o considering the gravity constraint	Proposed method
Case A1	2.4654	2.2266	2.4644	2.2256
Case A2	0.7122	0.7019	0.7098	0.6994
Case A3	0.4222	0.4207	0.4181	0.4166
Case B1	2.4955	2.2396	2.4945	2.2386
Case B2	0.8015	0.7903	0.7993	0.7881
Case B3	0.5602	0.5588	0.5571	0.5557

After demonstrating that the proposed method achieves greater accuracy than the sensor reading method, it is necessary to compare the significance of taking the gravity constraint into account in the proposed scheme. Comparing the results of the two methods from **Table III**, the *rms* of gravity and linear acceleration errors of the method without considering the gravity constraint are always greater than the errors of the proposed method. Comparing equation (25) with equation (19), the term with gravity constraint, $-k_e\left\|\hat{\mathbf{g}}\right\|^{-1}e\hat{\mathbf{g}}$, is excluded in equation (25). However, the results from **Table III** show that it is crucial to include the term with gravity constraint. In equation (19), the role of the term with gravity constraint, $-k_e\left\|\hat{\mathbf{g}}\right\|^{-1}e\hat{\mathbf{g}}$, is to adjust the estimate of the gravity vector, $\hat{\mathbf{g}}$, in the opposite direction of $e\hat{\mathbf{g}}$. This adjustment reduces the magnitude of e, where e is the error, $\left\|\hat{\mathbf{g}}\right\|-\mathrm{g}_0$, defined in equation (10). In other words, the term with gravity constraint, $-k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{\mathbf{g}}$, will always be in parallel with $\hat{\mathbf{g}}$, and results in the conclusion of reducing $|e|$ whenever $|e| \neq 0$. Thus, the result of $\hat{\mathbf{g}}$ in equation (19) provides a more accurate estimation as compared to the result of $\hat{\mathbf{g}}$ in equation (25). It is essential to include the term, $-k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{\mathbf{g}}$, in the estimation of the gravity vector.

Nevertheless, the term, $-k_g(\hat{\mathbf{g}} - \mathbf{a})$, also serves as a key factor in enhancing the accuracy of the estimation of the gravity vector. In equation (19), when $\hat{g}_i > a_i$, $i = x, y, z$, then $-k_g(\hat{g}_i - a_i) < 0$. As a consequence, $\dot{\hat{g}}_i$ is negative, making $\hat{g}_i$ approaches a_i. Therefore, the estimate of the gravity vector, $\hat{\mathbf{g}}$, will be adjusted toward $\mathbf{a}$. Based on that, k_g can be denoted as the coefficient that determines the adjusting rate of the gravity vector.

VI. Evaluation

This section will focus on the limitation and the weakness of this study. The *Assumption* presented in the **Methodology** states that the time variation of the gravity vector is small compared to the time variation of the object's linear acceleration. This is the limitation of this study. Because when the gravity vector and the linear acceleration have similar time variations, it is difficult to distinguish them from the accelerometer's outputs according to the relation (6). This limitation can be also observed in the simulation results with various frequencies of the object's linear acceleration. In the simulation study, the object's rotation frequency is fixed to 0.1 Hz. As shown in **Table I**, when the frequency of an object's linear acceleration increases, the difference between this frequency and 0.1 Hz increases, and the estimation errors of tilt decrease. Likewise, as shown in **Table II**, when the frequency of an object's linear acceleration increases, the estimation errors of both the gravity vector and the linear acceleration decrease. On the contrary, when the frequency of the linear acceleration is close to the object's rotation frequency, the estimation errors become relatively significant, which indicates the weakness of this study.

VII. Conclusion

In this study, the research question is about the estimation of the gravity vector. A salient feature of the proposed method is the consideration of the magnitude constraint on the gravity vector. Asymptotic convergence of the estimation error is ensured under an assumption. Two applications of this gravity estimation are the tilt estimation and the estimation of the linear acceleration of an object. The analysis shows that the angle errors of the proposed method are smaller than those of the sensor reading method. This result illustrates the significance of the identification of the gravity vector by the proposed method compared with the sensor reading method. The analysis also shows that the consideration of the magnitude constraint on the gravity vector in the estimation process is advantageous. Therefore, it would be a beneficial estimation of the gravity vector if the proposed method is used.

Referring back to the traffic baton, the state of the baton can be determined based on the proposed method and used to control the desired light-on or -off state of the baton. Therefore, the design of the traffic baton can then be improved as the estimation of the gravity vector in the proposed method becomes more accurate than the conventional method. It paved the way for further exploration in the design of the traffic baton that could impact society. By observing the phenomena in the real-world and addressing the challenges in society, this study lays the mathematical groundwork for further advancements in this field and within our community.

Bibliography

Craig, J. J. (2005). *Introduction to Robotics: Mechanics and Control*. 3rd Edition, Pearson Prentice Hall, Upper Saddle River.

DiSalle, R. (2002). *Space and Time: Inertial Frames.* Stanford.edu. https://plato.stanford.edu/entries/ spacetime-iframes/

Hemminki, S., Nurmi, P., & Tarkoma, S. (2014). Gravity and linear acceleration estimation on mobile devices. In *Proceedings of the 11th International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services*, 50-59.

Hodson, T. O. (2022). Root mean square error (RMSE) or mean absolute error (MAE): When to use them or not. *Geoscientific Model Development Discussions*, *2022*, 1-10.

Hover, F.S. & Triantafyllou, M.S. (2010). *System Design for Uncertainty*. Cambridge, MA: MIT Center for Ocean Engineering.

Pedley, M. (2013). Tilt sensing using a three-axis accelerometer. *Freescale Semiconductor Application Note*, AN3461.

Luczak, S., Waldemar Oleksiuk, & Maciej Bodnicki. (2006). Sensing tilt with MEMS Accelerometers. *IEEE Sensors Journal*, *6*(6), 1669–1675.

Luinge, H.J. & Veltink, P.H. (2004). Inclination measurement of human movement using a 3-D accelerometer with autocalibration. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 12(1), 112-121.

Teschl, G. (2012). *Ordinary Differential Equations and Dynamical Systems*. Providence: American Mathematical Society. ISBN 978-0-8218-8328-0.

Zohra Aziz Ali Manjiyani, Renju Thomas Jacob, Keerthan Kumar R, Babu Varghese (2018). Development of MEMS based 3-axis accelerometer for hand movement monitoring. *Int. J. Sci. Res. Publ.*, 4(2), http://www.ijsrp.org/research-paper-0214.php?rp=P262293.

Video links:

1. https://drive.google.com/file/d/1JK5EtD23Do3bWuBD0vWpP90PvV7z3u3F/view?usp=drive_link
2. https://drive.google.com/file/d/1Ey3_C_Xw-w32-hzcETgucyS1d5EfRx85/view?usp=drive_link

Appendix A

Solve $\|\hat{\mathbf{g}}\|^2 = \dfrac{\mathrm{g}_0^2}{\left[1+k_g^{-1}k_e\|\hat{\mathbf{g}}\|^{-1}\left(\|\hat{\mathbf{g}}\|-\mathrm{g}_0\right)\right]^2}$ for $\|\hat{\mathbf{g}}\|$. Expanding the original equation, one has:

$$\|\hat{\mathbf{g}}\|^2 + 2k_g^{-1}k_e\|\hat{\mathbf{g}}\|\left(\|\hat{\mathbf{g}}\|-\mathrm{g}_0\right) + \left[k_g^{-1}k_e\left(\|\hat{\mathbf{g}}\|-\mathrm{g}_0\right)\right]^2 = \mathrm{g}_0^2. \tag{A1}$$

Rearranging this equation gives:

$$\left(1+2k_g^{-1}k_e + (k_g^{-1}k_e)^2\right)\|\hat{\mathbf{g}}\|^2 - \left(2k_g^{-1}k_e + 2(k_g^{-1}k_e)^2\right)\mathrm{g}_0\|\hat{\mathbf{g}}\| + \left((k_g^{-1}k_e)^2 - 1\right)\mathrm{g}_0^2 = 0, \text{ or}$$

$$\left(1+k_g^{-1}k_e\right)^2\|\hat{\mathbf{g}}\|^2 - 2k_g^{-1}k_e\left(1+k_g^{-1}k_e\right)\mathrm{g}_0\|\hat{\mathbf{g}}\| + \left((k_g^{-1}k_e)^2 - 1\right)\mathrm{g}_0^2 = 0. \tag{A2}$$

The left-hand side of equation (A2) can be factorized, giving:

$$\left[\|\hat{\mathbf{g}}\| - \mathrm{g}_0\right]\times\left[\left(1+k_g^{-1}k_e\right)^2\|\hat{\mathbf{g}}\| - \left((k_g^{-1}k_e)^2-1\right)\mathrm{g}_0\right] = 0. \tag{A3}$$

Hence, the possible solutions for $\|\hat{\mathbf{g}}\|$ are g_0 and $\dfrac{(k_g^{-1}k_e)^2-1}{\left(1+k_g^{-1}k_e\right)^2}\mathrm{g}_0$. If $k_g > k_e > 0$, then the second possible solution is negative, which cannot be a solution for $\|\hat{\mathbf{g}}\|$. Because $\|\hat{\mathbf{g}}\|$ is always greater than or equal to zero. Thus, the unique solution for $\|\hat{\mathbf{g}}\|$ is g_0 when $k_g > k_e > 0$.

As shown in the first paragraph, solving (18) for $\|\hat{\mathbf{g}}\|$ gives $\|\hat{\mathbf{g}}\| = \mathrm{g}_0$ and $\dfrac{(k_g^{-1}k_e)^2-1}{\left(1+k_g^{-1}k_e\right)^2}\mathrm{g}_0$. Choose the parameters, k_g and k_e, to satisfy $k_g > k_e > 0$. Since $\|\hat{\mathbf{g}}\|$ must be greater than or equal to zero, the unique solution is $\|\hat{\mathbf{g}}\| = \mathrm{g}_0$, that is, $e = 0$. This together with (13) gives $\hat{g}_x = g_x$, $\hat{g}_y = g_y$, and $\hat{g}_z = g_z$. Hence, the largest invariant set inside *S* is $(\hat{g}_x, \hat{g}_y, \hat{g}_z) = (g_x, g_y, g_z)$. Based on the Krasovskii-Lasalle invariance principle (Teschl, G., 2012), it is concluded that $\hat{\mathbf{g}}$ approaches $\mathbf{g}$ asymptotically, which means that the estimate of the gravity vector converges to the actual gravity vector as time approaches infinity.

Appendix B

In the proposed algorithm, the derivative of $\mathbf{g}$ with respect to time is assumed to be zero, that is, $\dot{\mathbf{g}} = \mathbf{0}$. To release this assumption, a tri-axial gyroscope can be introduced. The gyroscope's output is denoted as $\boldsymbol{\omega}=\begin{bmatrix} p & q & r \end{bmatrix}^{\mathrm{T}}$, in which p, q, and r respectively represent the angular velocity around the x-, y-, and z-axes. The derivative of $\mathbf{g}$ can obtained by transforming $\mathbf{g}$ with $\boldsymbol{\omega}$, that is: (Hover, F.S. & Triantafyllou, M.S., 2010)

$$\dot{\mathbf{g}} = \begin{bmatrix} \dot{g}_x \\ \dot{g}_y \\ \dot{g}_z \end{bmatrix} = \begin{bmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{bmatrix} \begin{bmatrix} g_x \\ g_y \\ g_z \end{bmatrix} = \begin{bmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{bmatrix} \mathbf{g}. \tag{B1}$$

In practice, $\mathbf{g}$ is usually unknown so (B1) is implemented as:

$$\dot{\hat{\mathbf{g}}} = \begin{bmatrix} \dot{\hat{g}}_x \\ \dot{\hat{g}}_y \\ \dot{\hat{g}}_z \end{bmatrix} = \begin{bmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{bmatrix} \begin{bmatrix} \hat{g}_x \\ \hat{g}_y \\ \hat{g}_z \end{bmatrix} = \begin{bmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{bmatrix} \hat{\mathbf{g}}, \tag{B2}$$

in which $\dot{\hat{\mathbf{g}}}$ denotes an estimate of $\dot{\mathbf{g}}$, and $\dot{\hat{g}}_{\mathrm{i}}$ is an estimate of $\dot{g}_{\mathrm{i}}$ for $\mathrm{i} = x, y, z$. The estimation algorithm is then modified to:

$$\begin{aligned} \dot{\hat{g}}_x &= \dot{\hat{g}}_x - k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_x - k_g(\hat{g}_x - a_x), \\ \dot{\hat{g}}_y &= \dot{\hat{g}}_y - k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_y - k_g(\hat{g}_y - a_y), \\ \dot{\hat{g}}_z &= \dot{\hat{g}}_z - k_e \|\hat{\mathbf{g}}\|^{-1} e\hat{g}_z - k_g(\hat{g}_z - a_z), \end{aligned} \tag{B3}$$

in which $\dot{\hat{g}}_{\mathrm{i}}$ for $\mathrm{i} = x, y, z$ are obtained from (B2). The assumption is released by introducing a tri-axial gyroscope, and the estimation algorithm (B3) aims to reduce the estimation lag under the rotational motion of an object.